IP-SAKTI Sahayak

Complete System Architecture, Features & Data Workflow | Smart India Hackathon

The problem

AYUSH startups, Ayurvedic practitioners and researchers cannot easily tell whether a herbal formulation is **classical** (largely unpatentable and defended by traditional-knowledge records), has genuine **patent potential**, or triggers **Access-and-Benefit-Sharing (ABS)** obligations under the Biological Diversity regime. Getting this wrong is expensive — and the law itself is a moving target (2023–24 biodiversity & patent rules, the WIPO GRATK Treaty). Expert legal counsel is costly and slow.

Our solution

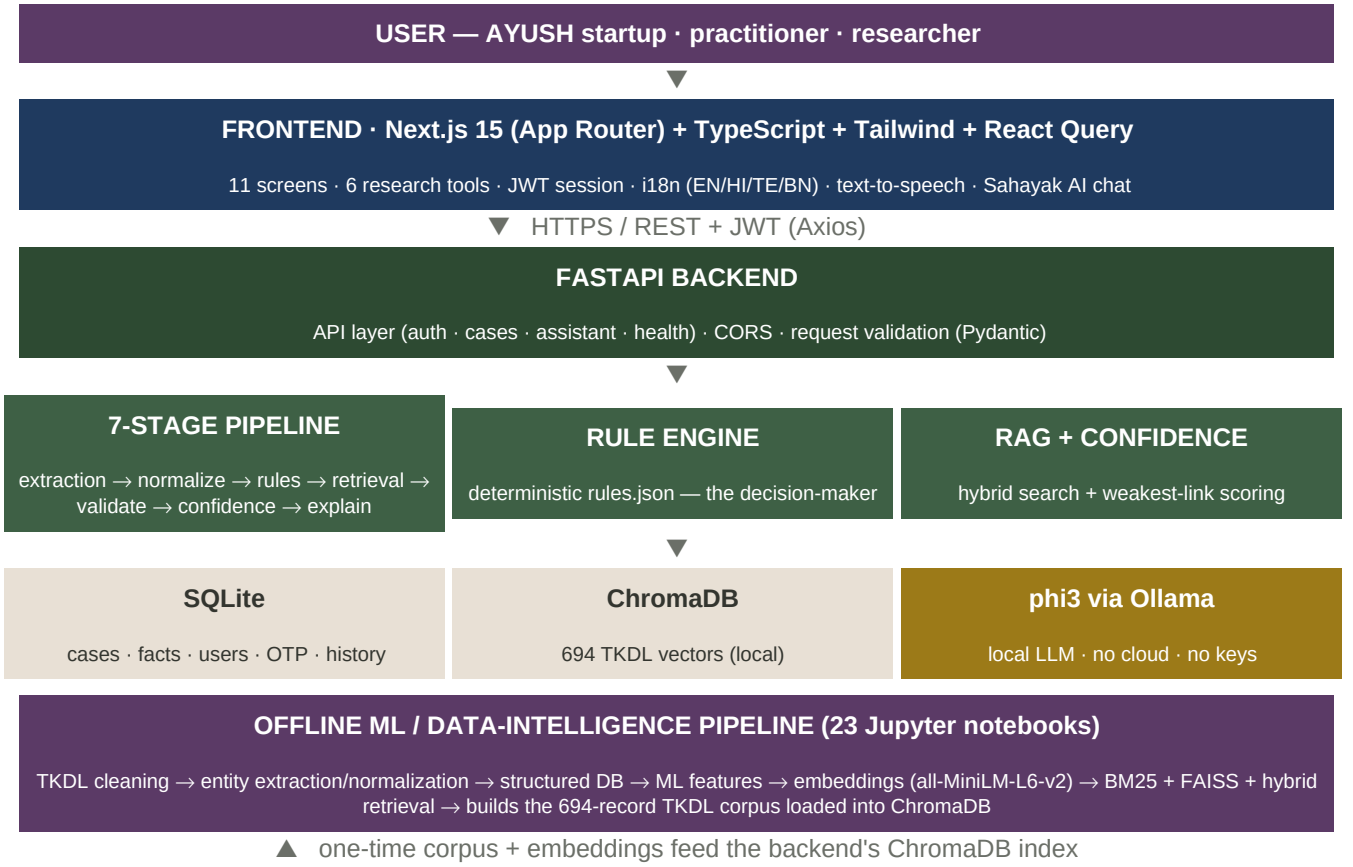
IP-SAKTI Sahayak is an **evidence-grounded decision-support system**. A user describes a formulation; the system extracts structured facts, a **deterministic rule engine** classifies it and names the exact statute, real **TKDL prior-art evidence** is retrieved and validated, a **multi-dimensional confidence** score is assigned, and the user gets a fully **explainable** result — or the system abstains and escalates to a human. Everything runs on a **local LLM (phi3 via Ollama)**: no cloud, no API keys, fully private and auditable.

What makes it different (for the judges)

Deterministic, not a guess The legal conclusion comes from an auditable rule engine — the AI is structurally forbidden from writing the classification.	**Grounded in real evidence** Every answer is backed by cited records from a 694-formulation TKDL corpus, not hallucinated text.	**Private & offline** Runs entirely on a local phi3 model through Ollama — suitable for sensitive IP with zero cloud dependency.
Honest about uncertainty Weakest-link confidence; when evidence is thin the system abstains and routes to a human reviewer.	**Temporal & auditable** As-of-date reasoning on statutes, full case history & versioning, and a step-by-step trace for every decision.	**Accessible** Multilingual UI (EN/HI/TE/BN), text-to-speech, and an Ayurveda-aware grounded chatbot.

1 · System architecture

Three cooperating subsystems — a Next.js frontend, a FastAPI backend, and an offline ML/data pipeline — over a local data + AI layer.



Design principle: RAG *supplies evidence*; the rule engine *makes the decision*; the LLM only *reads* free text into facts and *explains* results. This separation is what makes every output auditable, repeatable and defensible.

2 · Feature catalogue

Every capability in the system, grouped by area.

Authentication & accounts

FEATURE	WHAT IT DOES
Register + email OTP	Sign up, verify via a 6-digit email OTP (dev fallback prints OTP to server log).
Password & OTP login	JWT-based sessions (bcrypt-hashed passwords); login by password or email OTP.
Profile & preferences	Edit name/role/address and default jurisdiction; session auto-expires and re-auth is enforced.

Six research tools (specialised intake & result surfaces)

TOOL	PURPOSE
Formulation Classifier	Classify a formulation (classical / proprietary / new-drug pathway) from ingredients, source texts, processing, claims and modifications.
ABS Compliance	Assess biodiversity Access-and-Benefit-Sharing obligations for biological resources and their access context.
TKDL / Prior Art	Search public prior-art candidates; clearly separates public pointers from restricted TKDL content.
IP Navigator	Map applicable IP routes — Patent, GI, Trademark, Design, Copyright, Plant Variety.
Regulatory Pathway	Surface the regulatory route (D&C / AYUSH / FSSAI), claim restrictions and labelling review items.
Ask Sahayak	A grounded question-and-answer surface with cited evidence and a confidence indicator.

Core analysis & intelligence

FEATURE	WHAT IT DOES
LLM fact extraction	phi3 (JSON mode) turns free text into 5 structured facts with source spans + confidence; deterministic keyword fallback if the model is offline.
Human fact verification	The user confirms/edits every fact before analysis — the cheapest guard against a bad AI read.
Deterministic rule engine	Evaluates confirmed facts against rules.json with as-of-date filtering, priority ordering and conflict detection.
Hybrid RAG retrieval	BM25 keyword + MiniLM vector search over 694 TKDL records for the most relevant prior art.
Citation validation	Keeps only evidence whose retrieval relevance clears a calibrated threshold.
Multi-dimensional confidence	5 scores, weakest-link (min) gated, mapped to HIGH / MEDIUM / LOW / INSUFFICIENT.
Abstention & escalation	When evidence/facts are insufficient the system abstains and offers one-click human review.
Explainability trace	A step-by-step stepper: facts → rule → citation → evidence → confidence.
Sahayak AI chatbot	Greeting-aware, with an LLM Ayurveda gatekeeper; answers are grounded in retrieved TKDL evidence.

Case management, output & accessibility

FEATURE	WHAT IT DOES
Dashboard & analytics	Case pipeline stats and a classification mix chart across all cases.
Case history & versioning	Full audit trail; re-analysis creates new versions without overwriting prior ones.
Archive + PDF export	Search/filter/sort past cases; export the filtered set as a PDF.
PDF case report	One-click formatted PDF: metadata, classification, confidence table, evidence and disclaimer.
Source library	Catalogue of corpus categories with version/access metadata.
Multilingual + TTS	English / Hindi / Telugu / Bengali UI and text-to-speech read-aloud for accessibility.

3 · End-to-end data workflow

The lifecycle of a single case, from description to exportable report.

1. Create case

User picks a tool and submits a formulation brief. Stored in SQLite (status PENDING).

2. Extract facts

phi3 reads the description into 5 structured yes/no facts with source spans.

3. Verify facts

User reviews and confirms/edits — facts marked CONFIRMED. Human-in-the-loop.

4. Normalize

Ingredient names canonicalised via an alias map before rule matching.

5. Rule engine

Confirmed facts + as-of date → classification + the exact rule and statute that fired.

6. Retrieve evidence

Hybrid BM25 + vector search returns ranked TKDL prior-art records.

7. Validate + score

Citations checked for relevance; 5-dimension weakest-link confidence → band.

8. Explain + persist

Trace assembled; result, history & version written back to the case.

HIGH / MEDIUM → RESULT

classification, evidence, next steps

LOW → CLARIFY

ask for more facts

INSUFFICIENT → ABSTAIN

escalate to a human reviewer

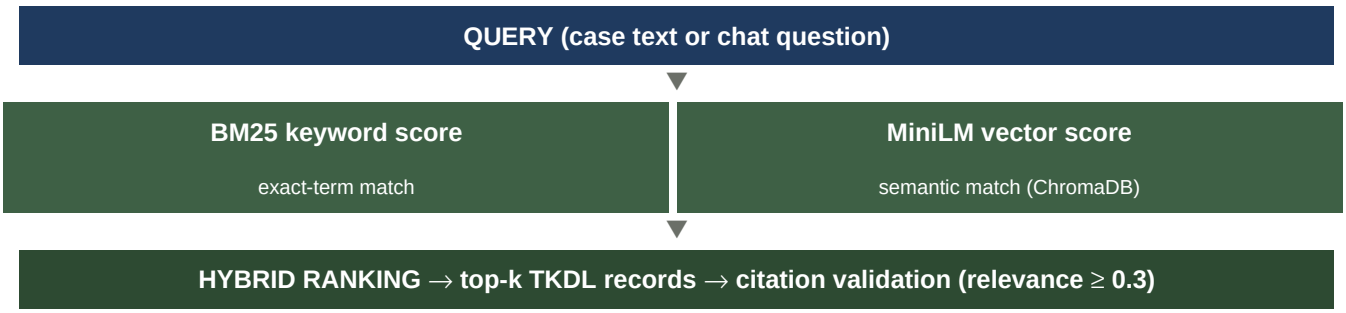
Then: **Explainability** stepper, **PDF report** export, **case history** / re-analysis versioning, and **escalation** — all reading from the single persisted case record.

The analysis pipeline — stage contracts

STAGE	INPUT → OUTPUT	TECH
1 Fact extraction	description → Fact[] (boolean + span + confidence)	`phi3 / Ollama, JSON mode`
2 Normalization	Fact[] → canonicalised Fact[]	`alias dictionary`
3 Rule engine	Fact[] + date → classification + fired rule(s)	`plain Python, rules.json`
4 Retrieval	query → ranked EvidenceChunk[]	`ChromaDB + BM25 hybrid`
5 Validation	chunks → validated evidence (relevance ≥ 0.3)	`relevance threshold`
6 Confidence	scores → 5-dim breakdown + band	`weakest-link (min)`
7 Explainability	analysis → UI trace (no new computation)	`trace assembler`

4 · Retrieval (RAG) & the grounded chatbot

The corpus is **694 real TKDL formulation records** (Ayurveda / Unani / Siddha), embedded once with **sentence-transformers all-MiniLM-L6-v2** (384-dim) and stored in **ChromaDB**. Queries are answered by a **hybrid ranking** — BM25 keyword score combined with cosine vector similarity — so results are strong on both meaning and exact terminology.



Retrieval quality (evaluated offline on 1,540 source-grounded queries)

MRR	Hit@1	Hit@3	Hit@5	Hit@10
0.82	0.77	0.86	0.88	0.89

Sahayak AI chatbot — intent routing

Every chat message is routed before any heavy work, keeping answers on-topic and fast:

GREETING	AYURVEDA	NOT AYURVEDA
short friendly reply (no retrieval, no LLM)	LLM gatekeeper passes → RAG + grounded answer	polite redirect to in-scope topics

The gatekeeper is an LLM classifier (AYURVEDA / NOT_AYURVEDA) that understands Ayurvedic principles, herbs, diet, treatments and classical texts as in-scope — far more flexible than keyword matching. Answers cite the retrieved records inline and never fabricate sources.

Confidence model

Five independent scores (0–1) are combined by a **weakest-link (minimum) gate** — one weak link caps the overall score, preventing over-confident conclusions:

DIMENSION	MEANING
fact_conf	quality of the extracted facts
rule_fit_conf	did a rule cleanly fire
evidence_conf	strength of retrieved prior art
citation_conf	did evidence pass validation
overall_conf = min(all)	HIGH ≥ 0.80 · MEDIUM ≥ 0.50 · LOW ≥ 0.25 · else INSUFFICIENT (abstain)

5 · API surface

METHOD	ENDPOINT	PURPOSE
POST	`/auth/signup · /verify-otp · /login · /request-otp`	Register, verify OTP, sign in (JWT).
GET/PATCH	`/auth/me`	Read / update the signed-in profile.
POST/GET	`/cases · /cases/{id}`	Create, list and open cases.
POST	`/cases/{id}/facts`	Extract facts (AI) or persist confirmed facts.
POST	`/cases/{id}/analyze`	Run the 7-stage pipeline.
GET	`/cases/{id}/explain`	Return the reasoning trace.
GET	`/cases/{id}/report`	Download a formatted PDF report.
POST	`/cases/{id}/escalate · /export-pdf · /upload`	Escalate, export archive PDF, extract text from an uploaded PDF.
POST	`/assistant/chat`	Grounded Ayurveda Q&A (phi3 + TKDL evidence).

6 · Data model (SQLite)

TABLE	KEY FIELDS / ROLE
FormulationSubmission	The case: submission_id, title, description, jurisdiction, as-of date, tool, status, and full result_json (classification, confidence, evidence, history, versions).
ExtractedFact	Per-fact rows: field, value, normalized_value, status (EXTRACTED/CONFIRMED), confidence, source_span.
User	email, bcrypt password hash, verification flag, profile, default jurisdiction.
OTPRecord	Time-limited email OTPs for verification / OTP login.
TKDLRecord	Structured TKDL entry schema used alongside the vector corpus.

7 · Technology stack

LAYER	TECHNOLOGY
Frontend	Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS, TanStack React Query, Axios, Recharts, Sonner.
Backend	FastAPI, Pydantic, SQLAlchemy, Uvicorn; JWT (python-jose) + bcrypt; fastapi-mail; reportlab (PDF); pypdf.
AI / RAG	phi3 via Ollama (local); sentence-transformers all-MiniLM-L6-v2; ChromaDB; rank-bm25.
Data / ML	SQLite; 23-notebook offline pipeline producing the TKDL corpus, embeddings, FAISS index & features.
Quality	pytest unit + integration suite (rule engine, confidence, normalization, temporal logic, citation validation).

Honest scope. This is a prototype for **research support, not legal advice**. The classification is made by a transparent rule engine and the retrieval foundation is independently evaluated; results should be confirmed with a qualified professional. The system is explicitly designed to **abstain** rather than over-claim when evidence is insufficient.